

ATES-metal: User Manual

Scope. This manual describes how to set up, run, and analyze models with ATES-metal, the in-house extension of the ATES (ATmospheric EScape) code to trace metals, metal-line cooling, a runtime opacity dispatcher, the Roche tidal potential, and excited-hydrogen / Ly α physics. It documents every input parameter (with a brief physical meaning), every output file (column by column, with units), the analysis tools, and a set of worked examples. The detailed physics derivations live in the companion notes (Update_ATES_early_phase.tex, H α transmission.tex, Update_ATES.tex); this document is the practical user guide.

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1. Introduction

1.1 What the code does

ATES-metal solves the time-dependent, spherically-symmetric (1-D) radiation–hydrodynamics of an irradiated planetary upper atmosphere and the hydrodynamic wind it drives (“atmospheric escape”). It integrates the Euler equations for mass, momentum, and energy of a single fluid,

$$\partial_t \rho + \frac{1}{r^2} \partial_r (r^2 \rho v) = 0, \quad \partial_t (\rho v) + \frac{1}{r^2} \partial_r (r^2 \rho v^2) + \partial_r p = -\rho \partial_r \Phi,$$

$$\partial_t E + \frac{1}{r^2} \partial_r (r^2 v (E + p)) = -\rho v \partial_r \Phi + \mathcal{H} - \mathcal{C},$$

to a steady state, where Φ is the gravitational (optionally Roche) potential (§1.3), $\mathcal{H}$ is the XUV photoionization heating and $\mathcal{C}$ the radiative cooling. Coupled to the hydrodynamics, at every step it solves the *ionization equilibrium* of H, He, and the trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) with a Newton (MINPACK) solver, including photoionization, collisional ionization, radiative + dielectronic recombination, and charge exchange. The converged run yields the radial profiles of density, velocity, temperature, and the ionization state, plus the steady-state *mass-loss rate* $\dot{M}$. A post-processing pass adds advection to the ionization balance, and the Python tool TPM.py turns a converged model into transit *transmission spectra* (He 10830, Ly α , H α /H β , and the metal resonance lines Mg II, Ca II, Na I).

1.2 The lower boundary, radii, and normalization

The computational domain starts at a **lower boundary** placed at the 1 μ bar pressure level, which defines $r = 1$ and the base radius R_p (the input Planet radius). The base number density n_0 and temperature T_0 (the equilibrium temperature) are set there. Internally the code is non-dimensional: lengths in R_p , velocities in $v_0 = \sqrt{k_B T_0 / \mu}$, densities

in n_0 , etc. All output files are written back in cgs (see §4. for the exact normalization of each column). The key dimensionless number is the *Jeans / escape parameter* $b_0 = GM_p\mu/(k_B T_0 R_p)$, the depth of the gravitational well in units of the thermal energy; small b_0 means an easily-escaping (or tidally unbound) atmosphere.

1.3 The gravitational potential

The potential Φ in the momentum and energy equations is selected by Domain mode (Table 3). The code writes it in the dimensionless units introduced above (lengths in R_p , potential in $v_0^2 = k_B T_0 / \mu$), so it is fixed by just three numbers: the Jeans parameter $b_0 = GM_p\mu/(k_B T_0 R_p)$, the stellar-to-planet mass ratio $q = M_*/M_p$, and the orbital separation $\tilde{a} = a/R_p$.

Spherical mode (Domain mode: Spherical). Only the planet's own gravity is kept, so the domain can extend far past the Hill radius:

$$\phi(r) = -\frac{b_0}{r}.$$

Roche mode (default). The full co-rotating Roche potential, evaluated along the star–planet (substellar) line $x = r$, adds the stellar tidal attraction and the centrifugal term of the synchronously-rotating frame:

$$\phi(r) = -\frac{b_0}{r} - \frac{b_0 q}{\tilde{a} - r} - \frac{b_0(1+q)}{2\tilde{a}^3} \left(\frac{q\tilde{a}}{1+q} - r \right)^2,$$

where the second term is the star's gravity at distance $\tilde{a} - r$ and the third is the centrifugal potential about the center of mass at $x_{\text{cm}} = q\tilde{a}/(1+q)$ with orbital frequency $\Omega^2 = G(M_p + M_*)/a^3$ (synchronous rotation). The effective gravity used by the hydrodynamics is $g(r) = -d\phi/dr$, with

$$\frac{d\phi}{dr} = \frac{b_0}{r^2} - \frac{b_0 q}{(\tilde{a} - r)^2} + \frac{b_0(1+q)}{\tilde{a}^3} \left(\frac{q\tilde{a}}{1+q} - r \right).$$

The inner Lagrange point L_1 is the root of $d\phi/dr = 0$, where planetary gravity, the stellar tide, and the centrifugal force cancel; in Roche mode the outer boundary is placed at this L_1 radius, beyond which gas is no longer bound to the planet.

Full 3-D form (transmission post-processing). The 3-D Roche-equipotential reconstruction in TPM.py / roche_recon.py (§5.2) needs the potential off-axis as well. With the substellar direction $\hat{x}$ pointing toward the star and (y, z) the sky plane,

$$\phi(x, y, z) = -\frac{b_0}{\sqrt{x^2 + y^2 + z^2}} - \frac{b_0 q}{\sqrt{(\tilde{a} - x)^2 + y^2 + z^2}} - \frac{b_0(1+q)}{2\tilde{a}^3} \left[(x_{\text{cm}} - x)^2 + y^2 \right],$$

which reduces to the substellar $\phi(r)$ above on the $+x$ axis. The centrifugal term involves only the rotation-plane coordinates (x, y) , not z , so surfaces of constant ϕ are triaxial (most extended toward L_1 , least along the spin axis z).

1.4 Workflow

A model run consists of:

1. Prepare `input.inp` (planet/star parameters, numerics, physics switches) — by hand or via the GUI; optionally `metals.inp` (trace metal abundances) and `opacity.inp` (opacity model).

2. Build and run `./ATES.x`; it reads the input files from the current directory and writes profiles to `output/`.
3. Inspect with `ATES_plots.py` (live profiles) and the example notebook; compute transmission spectra with `TPM.py`.

2. Installation and running

2.1 Requirements

A Fortran compiler (gfortran by default; Intel ifort/ix supported), make, and — for the GUI and analysis — Python 3 with tkinter, numpy, scipy, and matplotlib.

2.2 Build

ATES-metal uses an incremental Makefile; module dependencies are auto-generated, so only edited sources and their dependents recompile.

```
make                # build ./ATES.x with gfortran
make FC=ifort       # Intel ifort   (or FC=ifx for the LLVM compiler)
make clean          # remove .o/.mod (keeps ATES.x)
make distclean      # wipe build/ and ATES.x
```

2.3 Running

The driver script `run_ATES.sh` (1) launches a Python/Tk GUI that writes `input.inp`, (2) builds, and (3) runs `./ATES.x`:

```
chmod +x run_ATES.sh
./run_ATES.sh          # gfortran (GUI -> build -> run)
./run_ATES.sh --ifort  # Intel ifort   (--ifx for ifx)
```

You may equally skip the GUI: place a hand-written `input.inp` (and optional `metals.inp/opacity.inp`) in a `run` directory, ensure an `output/` subdirectory exists, and run `./ATES.x` there. Each modelled planet typically lives in its own subdirectory holding its `input.inp`, its `ATES.x`, and its `output*/` folders. Progress (iteration counter, the momentum-variation du , and the time-derivative norm dtu) is printed to the terminal; the run stops when du or dtu falls below its convergence threshold.

2.4 Convergence control

The time loop stops when *any* of the following criteria is met (defaults in `src/modules/init/parameters.f90`; most are runtime-settable, Table 3):

Parameter	Default	Stop condition
du_th	1×10^{-3}	momentum non-uniformity $du < du_th$
dtu_th	1×10^{-8}	time-derivative ∞ -norm $dtu < dtu_th$
stall_tol, N_stall	10^{-6} , 2000	du on a plateau for N_stall steps
count_max	10^6	hard iteration cap

The stall and count_max backstops guarantee termination even when the strict du/dtu thresholds are physically unreachable for a given setup (e.g. a transonic wind whose du floors at $\sim 1\%$). Each iteration prints count, du, dtu (and the windowed flux-level change when Level_tol: is set), and the run reports which criterion stopped it. The optional Force_start forces the first 1000 iterations regardless (useful when starting near a fixed point). The rationale and history of these settings are in Update_ATES_solver.

Caveat: du can stop a run prematurely. du measures only the *spatial spread* of pvr^2 in the wind window; it is blind to a uniform drift of the flux level, and on strongly irradiated planets it *oscillates*, dipping below 10^{-3} transiently long before the wind settles. A WASP-121b reference stopped this way was low by ~ 0.08 dex in $\dot{M}$ with the heating/cooling grossly out of balance (energy residual ~ 30). For production answers use the residual-based stop (Resid_tol:) or the Newton finish (Solver: Newton, §2.5); details in steady_solver_memo.

2.5 The steady-state Newton solver (Solver: Newton)

The finite-volume steady state can be solved directly — $F(Y) = 0$ for the hydro state, with the ionization equilibrium eliminated locally inside the residual — by a preconditioned Jacobian-free Newton–Krylov method (banded LAPACK preconditioner, frozen WENO weights, diagonal scaling, non-monotone line search; the smooth base valve Valve_eps: is required and auto-adopted). This reaches residual levels unattainable by marching at a small fraction of the cost: WASP-121b converges in 8 Newton iterations (~ 200 marching-steps of cost, vs. $> 35,000$ insufficient marching steps), HD189733b in 33 s where marching spent > 3 h without converging.

Two ways to use it:

- **Production:** add Solver: Newton to input.inp. The run marches (including the automatic two-stage PLM→WENO3) until the monitored steady residual drops below the switch threshold (default 5×10^{-2} ; the du -based stops are suspended), then the Newton solve finishes to Resid_tol (default 10^{-3}), and the standard outputs and post-processing follow.
- **From an existing state** (recommended for re-converging old runs): put the state in output/*_IC.txt, set Load IC? True, Valve_eps: 1.0e-4, Resid_tol: 1.0e-3, and run with the environment hook

```
ATES_PTC=1 ATES_PTC_JFNK=1 ATES_PTC_DTAU0=1.0 ./ATES.x
```

which solves from the loaded state, writes output/, and stops.

The warm start matters: Newton’s basin requires a state with $\max \|R\| \lesssim 5 \times 10^{-2}$ (a transient that merely *looks* flat in du is not enough). Do not raise the switch threshold: on WASP-121b a hand-off at $\|R\| = 0.09$ already stalls (and a cold-IC

start stalls at $\|R\| \sim 10$, while the default 5×10^{-2} converges in 16 iterations — the default is calibrated essentially at the basin edge (steady_solver_memo.pdf). Convergence is linear (frozen-weight refresh and the chemistry noise floor); on planets with a breathing base (HD189733b) the practical floor is $\|R\| \sim 3 \times 10^{-4}$. The Newton finish is also *safe to fail*: the solver tracks the best iterate seen, aborts once no new best residual has appeared for 15 outer iterations, and on failure the run prints `JFNK failed`, keeps the best iterate, and falls back to plain time-marching with the *du*-based stops re-armed — an unconverged Newton state is never accepted as the final answer. The associated diagnostic hooks (ATES_RESIDUAL, ATES_JAC_TEST, ATES_NEWTON_TEST, ...) are environment variables, collected in §2.6 (Table 1). Full development record and validation: steady_solver_memo.pdf. Figure 1 shows the production pipeline end-to-end on WASP-121b (Roche): the monitored residual during the marching warm-up (left; note that the *du*-based stop would have fired long before the residual is actually small) and the JFNK finish (right).

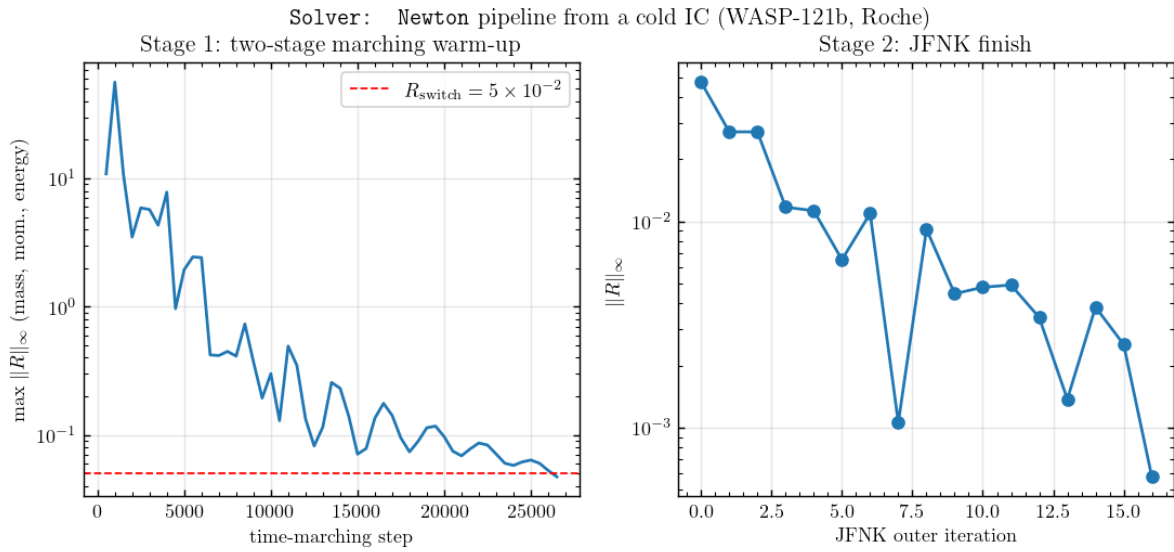


Figure 1: The Solver: Newton pipeline from a cold IC (WASP-121b, Roche). Left: maximum steady residual $\max \|R\|_\infty$ over the mass, momentum, and energy equations during the two-stage marching warm-up; the dashed line is the switch threshold $R_{\text{switch}} = 5 \times 10^{-2}$. Right: residual vs. JFNK outer iteration during the Newton finish, down to Resid tol (10^{-3}).

2.6 Environment-variable hooks (diagnostics and expert runs)

A small set of *environment variables* expose diagnostic and expert entry points without touching `input.inp` (Table 1). They are read once at startup; when unset they have strictly zero effect, so normal runs are unaffected. All of them are diagnostics or validation aids — the only one used in a production workflow is the ATES_PTC family, which implements the “Newton from an existing state” recipe of §2.5 (and `examples/inputs/04_newton_from_state/`). Typical usage:

```
ATES_RESIDUAL=1 ./ATES.x # rank a "converged" state
ATES_PTC=1 ATES_PTC_JFNK=1 ATES_PTC_DTAU0=1.0 ./ATES.x # re-converge it
```

Variable	Effect
ATES_DUMP_IC=1	After Load IC? True, dump the restored state to output/IC_dump.txt <i>before</i> the first ionization-equilibrium solve re-equilibrates the species fractions (verifies a snapshot/restart round-trip).
ATES_RESIDUAL=1	Evaluate the steady residual R of the loaded state (WENO3, boundary conditions applied first), print the per-equation norms (mass, momentum, energy), and stop. The criterion-independent way to rank candidate “converged” states (a premature-dip state shows $\ R\ _E \sim O(10)$ despite $du < 10^{-3}$).
ATES_NEWTON_TEST=1	Self-test of the Newton infrastructure (pack_U/unpack_U round-trip and the frozen residual against the ATES_RESIDUAL values); prints and stops.
ATES_JAC_TEST=1	Verify the colored-FD banded Jacobian against a directional finite difference $Jr \simeq [F(Y + \epsilon r) - F(Y)]/\epsilon$; a correct band layout matches to $\sim 10^{-6}$, a wrong one is $O(1)$ off. Prints and stops.
ATES_PTC=1	Solve the <i>loaded</i> state directly with the steady solver (no marching), write output/, and stop. Requires Load IC? True; set Valve eps: $1.0\text{e-}4$ and Resid tol: $1.0\text{e-}3$ (§2.5). Default engine: banded PTC.
ATES_PTC_JFNK=1	With ATES_PTC=1: use the full JFNK solver (GMRES on the matrix-free J) instead of the banded PTC — the recommended engine.
ATES_PTC_DTAU0= $\langle v \rangle$	With ATES_PTC=1: override the initial pseudo-time step $\Delta\tau_0$ (default: the CFL Δt of the loaded state). A large value (e.g. 1.0) probes the pure-Newton regime directly.
ATES_PTC_NFIX= $\langle n \rangle$	With ATES_PTC=1: anchor the first n physical cells (frozen-base experiment; solves the wind on top of a fixed base).
ATES_DIAG_BASE=1	Write output/base_diag.txt (r, n, v, T , heating, cooling of the first few cells over the first steps) to trace IC-startup transients at the base.
ATES_FORCE_HYBRD1=1	Force every ionization-equilibrium solve to use MINPACK hybrd1, skipping the analytic-Jacobian Newton: lets the same binary produce both solvers’ outputs for an A/B validation (Update_ATES_solver.pdf §6).

Table 1: Environment-variable hooks. All are read once at startup and have zero effect when unset. The four diagnostic tests (top block) and ATES_PTC runs stop after printing/writing; they do not march.

2.7 Metal cooling on/off and the comparison script

Metal-line cooling is active *iff* a `metals.inp` is present (§3.); remove or rename it for a metals-off run. To run the *same* planet both ways automatically:

```
./run_compare_metals.sh          # uses ./input.inp and ./metals.inp
```

This compiles once and runs the planet with metals on, then off, saving the two full results to `output_metals_on/` and `output_metals_off/` (each with a copy of the inputs used; the second run reuses the binary). The per-planet notebooks `metals_comparison*.ipynb` read the two folders and plot the temperature, velocity / mass-flux, H and metal ionization, and heating/cooling differences (cf. §7.).

3. Input files

3.1 `input.inp` — the main parameter file

`input.inp` is read *line by line in a fixed order* for the core block (Table 2), followed by an *optional keyword block* (Table 3) whose lines are matched by keyword and may be omitted (older files without them keep the default behavior). A few lines are *conditional*: the spectrum-property line depends on `Spectrum` type, and the energy-band / X-ray-luminosity lines depend on `Use only EUV?`.

Table 2: `input.inp` core block (fixed order). Conditional lines are marked (*).

Label	Value	Meaning
Planet name:	string	Label for the run (cosmetic).
Log10 lower boundary number density [cm ⁻³]:	real (log ₁₀)	log ₁₀ n_0 , the H number density at the 1 μ bar base.
Planet radius [R _J]:	real (R_J)	Base radius R_p (the 1 μ bar level); the grid radius r is in units of this.
Planet mass [M _J]:	real (M_J)	Planet mass M_p (sets gravity / the Jeans parameter b_0).
Equilibrium temperature [K]:	real (K)	Base temperature T_0 .
Orbital distance [AU]:	real (AU)	Star–planet separation a (sets the XUV flux $\propto a^{-2}$ and the tidal/Roche potential).
Escape radius [R _p]:	real (R_p)	Inner radius of the constant-momentum convergence window; must lie <i>inside</i> the domain ($< r_{\max}$). For L1-truncated Roche runs use a value below the lobe.
He/H number ratio:	real	Helium abundance $n_{\text{He}}/n_{\text{H}}$; > 0 enables He chemistry.

Table 2, continued

Label	Value	Meaning
2D approximate method:	see text	Day–night averaging of the XUV flux and $\dot{M}$: Rate/2 + Mdot/2 (dayside, $\xi = \frac{1}{2}$), Mdot/4 (full-sphere), Mdot, or alpha (then a trailing number sets the rate-correction a_τ).
Parent star mass [M_sun]:	real ($M_\odot$)	Stellar mass $M_\star$ (tidal potential, Roche-lobe size).
Spectrum type:	Power-law Monochromatic Load	Shape of the incident XUV SED.
(*) Power-law index: / Photon energy [eV]: / SED file:	real / real / path	Property of the chosen SED: power-law slope E^{PLind} , the monochromatic energy, or a numerical SED file. With metals, mind the required wavelength coverage (§3.5).
Use only EUV? True/False:	logical	True: EUV band only. False: include X-rays (adds the band’s third energy and an X-ray luminosity line).
(*) [E_low,E_mid,E_high] = [. . .]	reals (eV)	Band edges: two values (EUV only) or three (with X-rays).
(*) Log10 of X-ray luminosity [erg/s]: Log10 of EUV luminosity [erg/s]:	real (log ₁₀) real (log ₁₀)	Only if X-rays are included; log ₁₀ L_X . log ₁₀ L_{EUV} (with a , sets the incident XUV flux).
Grid type:	Uniform Stretched Mixed	Radial grid spacing. Mixed (fine near the base, stretched outward) is the usual choice.
Numerical flux:	HLLC ROE	Riemann solver for the hydrodynamic fluxes.
Reconstruction scheme:	PLM WENO3	Spatial reconstruction order.
Include He23S?:	logical	Include the metastable He 2 ³ S triplet (needed for the He 10830 line).
Load IC?:	logical	Restart from the previous run’s output/*_IC.txt (warm start).
Do only PP:	logical	Skip the time loop and only run the post-processing on existing output.

Table 2, continued

Label	Value	Meaning
Force start:	logical	Force the first 1000 iterations regardless of the convergence test (useful when the IC is close to a fixed point).

Table 3: input.inp optional keyword block (any line may be omitted; defaults restore the standard behavior).

Keyword	Value	Meaning
Domain mode:	Spherical	If present and Spherical: pure planetary potential $-b_0/r$, domain extended to Outer radius. Absent (default): full Roche potential truncated at the Hill/L1 radius (§1.3).
Outer radius [R_p]:	real > 1 (R_p)	Outer boundary in spherical mode (e.g. 10–15 R_p).
Stellar Teff [K]:	real (K)	Stellar T_{eff} for the diluted-blackbody Balmer continuum that photoionizes/heats H($n=2$). Enables the excited-H model when both this <i>and</i> Stellar radius are > 0 .
Stellar radius [R_sun]:	real ($M_{\odot}$ -radius)	Stellar radius (sets the Balmer-continuum dilution $R_{\star}/a$).
Jlya RT file:	path	Read an external $\bar{J}_{\text{Ly}\alpha}(r)$ profile to pump H($n=2$) (sets jlya_mode=1).
Jlya escape-prob:	True	Compute $\bar{J}_{\text{Ly}\alpha}$ in-line each step by escape-probability RT (jlya_mode=2).
Stellar Lya flux [erg/cm2/s]:	real	Incident stellar Ly α flux at the planet (the stellar source for jlya_mode=2).
Lya stellar halfwidth [km/s]:	real (km/s)	Stellar Ly α line half-width; controls how deep the trapped stellar beam penetrates (default 70).
Lya stellar boost [-]:	real	Bounded buildup cap for the trapped stellar Ly α beam (default 5).
Deexc heat:	True	Add collisional de-excitation heating of H($n=2$) to the energy budget.

Table 3, continued

Keyword	Value	Meaning
Transonic IC:	True	Use a transonic isothermal-wind initial condition (solved from the potential via the Bernoulli integral) instead of the hydrostatic IC; needed for deep-RLOF bases that do not launch from rest. Auto-enables Force start.
Newton solver:	logical (def. True)	Solve the ionization equilibrium with the analytic-Jacobian Newton method (with a MINPACK hybrd1 fallback); False reverts to the legacy hybrd1. Default is faster ($\sim 25\%$) and reproduces hybrd1 to $\sim 10^{-7}$.
Brent solver:	logical (def. True)	Solve the post-process energy equation with a bracketing (Brent) root-finder that selects the physical root; False reverts to legacy hybrd1 + a $2\times$ -band reject.
du_th [PLM,WENO3]:	two reals	Automatic two-stage reconstruction: run PLM until $du < \text{first value}$ (or a PLM stall), then switch to WENO3 and converge at $du < \text{second value}$ (e.g. 0.5×10^{-3}). Replaces the manual PLM $\rightarrow$ stop $\rightarrow$ Load IC+WENO3 restart of the official ATES recipe.
Stall [tol,N]:	real, int	Stall-detector override (relative du change and window; defaults 10^{-6} , 2000).
CFL:	real (def. 0.6)	CFL-number override.
Hot Parker IC:	real (K)	Warm-seed initial condition: stable hydrostatic density with a warm T /ionization/velocity overlay.
Level tol:	real	Mass-flux <i>level</i> -stability gate: a converged/stalled stop additionally requires the mean $ \rho v r^2 $ over the wind window to change by less than this (relative) across the last N_{stall} steps. Guards against stopping on a transient du dip while the flux level still drifts ($\leq 0 = \text{off}$, default; §2.5).

Table 3, continued

Keyword	Value	Meaning
Resid tol:	real	Residual-based convergence: stop when the steady residual $\max_k \max_j R_{j,k} / \max_j u_k $ falls below the value (monitored every 500 steps), <i>replacing</i> the <i>du</i> test (≤ 0 = off, default). <i>du</i> alone is blind to the energy imbalance (§2.5).
Time stepping:	Local	Per-cell pseudo-time steps (steady-state acceleration; not time-accurate). Reaches the deep state fast but leaves per-cell residuals; pair with a Newton finish.
Energy solver:	Explicit	Revert to the original forward-Euler source update (solver-isolation tests).
Valve eps:	real (code v)	Smooth base valve: ghost velocity $\frac{1}{2}(v + \sqrt{v^2 + \varepsilon^2})$ instead of $\max(v, 0)$. Differentiable at $v = 0$ (required by the Newton solver); the converged answer is insensitive to ε (10^{-4} vs. 10^{-5} tested identical). ≤ 0 = exact legacy valve (default).
Solver:	Newton [real]	Production steady-state solve: marching warm-up until the monitored steady residual $\max \ R\ $ drops below the optional value (default 5×10^{-2}), then the JFNK Newton finish to Resid tol (default 10^{-3}), then the standard final outputs and post-processing. Adopts Valve eps: 10^{-4} automatically if unset. See §2.5.

3.2 metals.inp — trace-metal abundances

`metals.inp` is *optional*: if absent, the run is metals-off (H/He only). If present, it lists the total elemental abundance n_X/n_H (summed over ionization stages; the equilibrium solver distributes each element over its stages) for any of C, N, O, Mg, Si, Ca, Na, K, S, Fe. Blank lines and # comments are ignored; any non-zero abundance turns metals on. Two control keys are also accepted: `pp_metals` (metal treatment in the advection post-process: 0=metal-free, 1=frozen at the converged equilibrium (default), 2=re-solve, not yet implemented) and `cx_full` (1=full Huang+2023 charge-exchange network; default metal-H only). Solar reference values (Asplund+2009):

```
# metals.inp -- solar n_X/n_H (Asplund+2009)
C 2.69e-4 N 6.76e-5 O 4.90e-4
Mg 3.98e-5 Si 3.24e-5 Ca 2.19e-6
Na 1.74e-6 K 1.07e-7 S 1.32e-5 Fe 3.16e-5
```

3.3 opacity.inp — opacity model (optional)

opacity.inp selects the bound-free opacity model via `OPACITY_MODEL = A|C|P|T`: **A** analytic (default, the standard ATES cross sections); **C** constant (threshold σ times the per-species factors `OPA_CONST_HI/HEI/HEII/HEIII`); **P** pressure-broadened (Robinson & Catling $f(p) = 1 + a(p/p_0)^n$ via `OPA_PB_FACTOR=a`, `OPA_PB_EXPONENT=n`, `OPA_PB_PIVOT=p_0`); **T** tabulated (per-species two-column .atesopa files, energy [eV] vs. σ [10^{-18} cm²], given by `OPA_FILE_HI` etc.). Missing keys keep their defaults, so an absent opacity.inp is equivalent to model A.

3.4 Excited hydrogen and the Ly α pumping field

The non-LTE $H(n=2)$ model (the metastable $2s$ + resonance $2p$ levels, Christie+2013) feeds two things: in `ATES.x` the $H(n=2)$ photoionization adds a Balmer proton source and photoelectric heating to the energy/ionization balance; in `TPM.py` the $n=2$ population sets the $H\alpha/H\beta$ transmission. The model is *off by default*; it switches on only when both `Stellar Teff` and `Stellar radius` are given (they supply the diluted-blackbody Balmer continuum that ionizes/heats $n=2$), so the standard run is unchanged.

The $2s/2p$ populations are driven by the **Ly α mean intensity** $\bar{J}_{\text{Ly}\alpha}(r)$, which pumps $1s \leftrightarrow 2p$. How $\bar{J}_{\text{Ly}\alpha}$ is obtained is selected by three mutually-exclusive options:

(mode 0) Parameterized — the default with no external input. If neither `Jlya RT file` nor `Jlya escape-prob` is present, $\bar{J}_{\text{Ly}\alpha}$ is the Huang et al. (2017, Eq. 6) estimate computed internally from the run’s own state,

$$\bar{J}_{\text{Ly}\alpha}(r) = \frac{0.1 F_{\text{LyC}}}{\Delta\nu_D(r)} \frac{1}{1 + \tau_{\text{Ly}\alpha}(r)},$$

where $\Delta\nu_D = \nu_0 \sqrt{2k_B T / \mu} / c$ is the local Doppler width, $\tau_{\text{Ly}\alpha}(r)$ is the *top-down* line-center Ly α optical depth integrated from the outside in through the HI column, and

$$F_{\text{LyC}} = \xi \frac{10^{L_{\text{EUV}}}}{4\pi a^2} \left(1 - e^{-\sigma_{\text{LyC}} N_{\text{HI}}}\right)$$

is the deposited Lyman-continuum flux (the incident EUV flux times the dayside factor ξ from 2D approximate method and the absorbed fraction set by the neutral-H column N_{HI}). This needs *no* external file — it is a self-contained, approximate “staging” field, adequate to get the coupling and signs right but not a true radiative-transfer solution.

(mode 1) External RT file — `Jlya RT file`: <path>. Read a tabulated $\bar{J}_{\text{Ly}\alpha}(r)$ profile (two columns, r/R_p and $\bar{J}$ in cgs) produced by a real Ly α radiative-transfer calculation (e.g. a Monte Carlo run); it is interpolated onto the grid and used directly.

(mode 2) In-line escape probability — `Jlya escape-prob`: True. Compute $\bar{J}_{\text{Ly}\alpha}$ every step from the current state by escape-probability radiative transfer (Neufeld/Harrington wing escape): a trapped *internal* field (recombination + collisional sources closed by the escape probability β) plus a trapped *stellar beam* set by `Stellar Ly $\alpha$  flux`

(with the penetration controlled by `Ly $\alpha$  stellar halfwidth` and the buildup capped by `Ly $\alpha$  stellar boost`). This is the most physical of the three but still an approximation; see `Update_ATES.tex` §12 for the full formulae.

`TPM.py` uses the *same* logic for its $\text{H}\alpha/\text{H}\beta$ $n=2$ population: if a `Jlya_file` is set it is read (mode 1), otherwise the mode-0 parameterized estimate above is used (with F_{LyC} rebuilt from the run's H I column and the stellar EUV). So a converged run with no $\text{Ly}\alpha$ input still yields $\text{H}\alpha/\text{H}\beta$ predictions, via the default field.

3.5 Spectral coverage for metal photoionization

Neutral metals have ionization thresholds *far below* the hydrogen Lyman limit ($13.6 \text{ eV} \equiv 912 \text{ \AA}$): the lowest is K I at 4.34 eV ($2856 \text{ \AA}$), followed by Na I, Ca I, Mg I, Fe I, Si I, S I, and C I (Table 4). To photoionize them at all, the modeled stellar spectrum must therefore reach these long (FUV/NUV) wavelengths — the EUV/X-ray band alone ($\lambda < 912 \text{ \AA}$, $E > 13.6 \text{ eV}$) leaves every neutral metal un-ionized and artificially abundant, corrupting both the ionization balance and the predicted metal lines. How far the spectrum reaches depends on the SED type (Spectrum type in Table 2):

Power-law / monochromatic (Power-law). The code builds its own energy grid. When low-IP metals are active it *automatically* appends a sub- 13.6 eV extension (20 log-spaced points) running from the H I edge down to the lowest ionization threshold among the active neutral metals, and evaluates the EUV power law there by *extrapolation* (the EUV normalization is anchored to L_{EUV} , which is defined over $[E_{\text{low}}, E_{\text{mid}}]$, then continued below E_{low}). No extra input is needed — but the FUV/NUV flux that ionizes the metals is just the EUV slope extended downward, which real stars do *not* follow (their flux rises steeply into the NUV). Treat power-law metal ionization as indicative; load a real SED for quantitative metal work. *Caveat:* the He 2^3S -triplet sub-grid and the metal sub-grid are mutually exclusive — a single power-law run cannot carry both.

Loaded SED (Load, SED file:). The file's own energy/wavelength grid *is* the radiative grid (read verbatim, no extrapolation). The SED file must therefore **extend to at least the lowest threshold of your active metals** — e.g. to $\approx 2860 \text{ \AA}$ if K is included, or to whichever neutral metal in Table 4 has the lowest threshold in the active set. A SED truncated at the Lyman limit ($912 \text{ \AA}$) delivers *zero* ionizing flux to Na I / K I / Ca I / Mg I / Fe I / Si I / C I / S I, so those stages stay spuriously neutral.

Either way, the energy bands set by $[E_{\text{low}}, E_{\text{mid}}, E_{\text{high}}]$ and the luminosities L_{EUV}/L_X still describe the *ionizing* (H/He-driving) part of the spectrum above 13.6 eV ; the metal extension is in addition to them.

4. Output files

All profiles are written to `output/` as plain text, one row per radial cell, in *cgs* units. Two flavors exist: the equilibrium files (`*.txt`, written during the run) and the advection-corrected files (`*_adv.txt`, written once at the end by the post-processor, which re-solves the ionization balance including advection of the wind). The `*_adv.txt` files are the ones the analysis tools read.

Table 4: Neutral-metal photoionization thresholds (first ionization stage of each modeled element), as E_{th} and the corresponding wavelength $\lambda_{\text{th}} = hc/E_{\text{th}}$. The stellar SED must extend to at least the longest λ_{th} among the *active* metals to ionize them. Values are the VFKY96/NIST thresholds used in the code (`species_table.f90`).

Species	E_{th} [eV]	λ_{th} [Å]	Note
K I	4.34	2856	lowest — sets the grid floor when K is present
Na I	5.14	2413	
Ca I	6.11	2028	
Mg I	7.65	1622	
Fe I	7.90	1569	
Si I	8.15	1521	
S I	10.36	1197	
Cl I	11.26	1101	
(H I)	13.60	912	Lyman limit (reference)
O I	13.62	910	$\simeq$ Lyman limit
N I	14.53	853	above the limit (already in the EUV band)

Caveat — the `_adv` profile at a “breathing” base. The advection-corrected (`*_adv.txt`) *temperature and ionization* would otherwise develop a spurious spike / sawtooth in the dense lower atmosphere (typically $r \lesssim 1.1\text{--}1.2R_p$ for strongly Roche-filling planets such as WASP-121b). Two compounding causes, both confined to the base, are responsible:

1. *A breathing, inflowing base (a 1-D limitation).* When the planet nearly fills its Roche lobe the 1-D wind is subsonic / only marginally launched at L_1 , so the dense lower atmosphere does not settle into a clean monotonic outflow but into a slowly recirculating state with small *negative* (inflow) velocities and a stagnation point ($v = 0$) near the wind base. This is intrinsic to the spherically-symmetric (1-D) geometry — removing it would require a 2-D/3-D model — and is not a numerical bug.
2. *A spurious hot root in the metal-cooled post-processor.* The post-processor re-solves the per-cell energy/ionization balance *including* advection, assuming an outflow (each cell takes the next-inner cell as its upwind state). In the dense base the non-monotonic metal line-cooling curve gives the energy equation a second, spurious *hot* root that the Newton (MINPACK) solver can land on; through the upwind coupling one bad cell then cascades outward, producing the sawtooth and an overshoot spike (`post_process_adv.f90`). A guard rejects solves that fall outside a band around the converged equilibrium temperature, but mild spikes *within* that band can survive.

The post-processor guards against cause 2 directly: wherever the flow is *not* a clean outflow ($v \leq 0$, the breathing inflow base) it skips the advection correction and keeps the converged equilibrium ionization and temperature (`post_process_adv.f90`). This removes the spurious spike and sawtooth at the source — it breaks the upwind cascade — while preserving the genuine few-percent `_adv` correction in the outflow above the stagnation point. Cause 1 (the breathing base itself) is intrinsic to the 1-D geometry and remains, but it is now represented there by the smooth, physical equilibrium solution. The shipped `_adv` profiles are therefore clean, and the temperature panel of Figure 6 plots

the `_adv` profile directly. (Velocity and density are *identical* in the `eq` and `_adv` files — the post-processor corrects only the ionization and temperature.)

4.1 Hydro_ioniz.txt / Hydro_ioniz_adv.txt

Thermodynamic profiles (7 columns). The dimensionless code variables are restored to cgs with the normalizations n_0 (base density), $v_0 = \sqrt{k_B T_0 / \mu}$, $p_0 = n_0 \mu v_0^2$, T_0 (base temperature), $q_0 = n_0 \mu v_0^3 / R_p$.

Col	Quantity	Unit	Note
1	r	R_p	radius (base = 1)
2	ρ (number density)	cm^{-3}	total density ($= \rho_{\text{code}} n_0$)
3	v	cm s^{-1}	radial velocity ($\times v_0$)
4	p	erg cm^{-3}	pressure ($\times p_0$)
5	T	K	temperature ($\times T_0$)
6	heating	$\text{erg cm}^{-3} \text{ s}^{-1}$	XUV photoionization heating ($\times q_0$)
7	cooling	$\text{erg cm}^{-3} \text{ s}^{-1}$	radiative cooling ($\times q_0$)

4.2 Ion_species.txt / Ion_species_adv.txt

Ion number densities [cm^{-3}] (34 columns). Column 1 is r ; columns 2–7 are the H/He ions; columns 8–34 are the 27 metal-ion stages, in this order:

`r ; HI HII HeI HeII HeIII HeI* ; CI CII CIII OI OII OIII NI NII NIII MgI MgII MgIII SiI SiII SiIII CaI
CaII CaIII NaI NaII KI KII SI SII FeI FeII FeIII`

(HeI* is the metastable 2^3S triplet.) The ionization fraction of any element is the stage density divided by the sum over that element's stages.

4.3 Cooling_breakdown.txt

Per-channel radiative cooling [$\text{erg cm}^{-3} \text{ s}^{-1}$] vs. radius (written for metals-on runs). Columns: 1= r , 2= T [K], 3= n_e [cm^{-3}], 4=total cooling; 5--8= the four H/He channels (recombination, collisional ionization, collisional excitation, bremsstrahlung); 9--35= the 27 metal-line cooling channels (same ion order as `Ion_species`). The channels sum to column 4. Useful for identifying the dominant coolant vs. altitude (cf. Huang+2023 Fig. 10).

4.4 Excited_H.txt

H($n=2$) diagnostic (written when the excited-H model is on, i.e. stellar T_{eff} and radius are set). 16 columns: 1 r , 2 T , 3 n_{HI} , 4 n_e , 5 $\bar{J}_{\text{Ly}\alpha}$, 6 n_{2s} , 7 n_{2p} , 8 Balmer proton source S_p , 9 photoelectric heating, 10 de-excitation heating, 11 $\tau_{\text{Ly}\alpha}$, 12--14 ground-state HI photoionization / collisional-ionization / recombination rates (for the proton-budget comparison), 15--16 the internal and stellar contributions to $\bar{J}_{\text{Ly}\alpha}$.

4.5 ATES.out and the mass-loss rate

ATES.out is a human-readable setup report (planet/star parameters, the Jeans parameter, composition, numerics) plus the final result. The **steady-state mass-loss rate** is $\dot{M} = 4\pi\rho v r^2$ evaluated near the outer boundary, corrected by the 2-D-approximation factor from 2D approximate method ($\text{Rate}/2 + \text{Mdot}/2 \Rightarrow \times \frac{1}{2}$, $\text{Mdot}/4 \Rightarrow \times \frac{1}{4}$). It is reported as $\log_{10}\dot{M}$ [g s^{-1}]; divide by M_p and multiply by 1 Gyr to express it in $M_p \text{ Gyr}^{-1}$. The helper `ates_io.mdot_log10 / mdot_Mp_per_Gyr` reproduce this.

5. Analysis tools

5.1 ATES_plots.py — live profile viewer

`python3 ATES_plots.py [--live N]` draws an 8-panel figure from `output/`: density, velocity, pressure, temperature, momentum $\log_{10}(\rho v r^2)$, the ion densities, and the H and He ionization fractions. With `--live N` it refreshes every N seconds (watch a run converge). If the `*_adv.txt` files exist they are overplotted (dashed) against the equilibrium profiles (solid). It also prints $\log_{10}\dot{M}$.

5.2 TPM.py — transmission spectra

TPM.py post-processes a converged run (`*_adv.txt + input.inp`) into transit transmission spectra and reports the line-center absorption depths (theoretical, instrument-convolved, and instrument+rotation-convolved) for **He 10830**, **Ly α** , **H α /H β** (from the non-LTE $n=2$ population, Christie+2013, pumped by the Ly α field of §3.4), and the metal resonance lines **Mg II 2796**, **Ca II K 3934**, **Na I D2 5890** (line-center and 4 Å-band depth). It supports two geometries via `geometry = 'spherical'` (Phase 5a) or `'triaxial'` (Phase 5b, the 3-D Roche-equipotential reconstruction in `roche_recon.py`). Set the planet/star parameters in its header (e.g. `R_star`, `rot_period`, `T_star`); these are *not* read from `input.inp` except R_p , M_p , a , L_{EUV} .

Step-by-step usage. TPM.py reads `output/Hydro_ioniz_adv.txt` and `output/Ion_species_adv.txt` from path (default `'.'`), so it is normally copied into the run directory and executed there:

1. *Ensure the *_adv.txt files exist.* A normal converged run writes them automatically. To post-process an existing state (e.g. a Newton solution produced by the ATES_PTC hook, which writes only the eq files): copy `Hydro_ioniz.txt` $\rightarrow$ `Hydro_ioniz_IC.txt` and likewise for `Ion_species`, set `Load_IC? True` and `Do only PP: True` in `input.inp`, and run `./ATES.x` once (seconds).
2. *Edit the header block* of the copied TPM.py: the stellar radius `R_star`, the rotation period `rot_period` (use the orbital period for a tidally locked planet), the stellar `T_star` (enables the Balmer-continuum $n=2$ photoionization; ≤ 0 disables), the instrument resolutions, and the output names `fig_name_hei/lya/ha/hb` (empty = do not save).
3. *Run* `MPLBACKEND=Agg python3 TPM.py` (the Agg backend suppresses the interactive windows on headless machines). The script saves the requested figures and prints the line-center absorption depths (theoretical /

instrument-convolved / +rotation) and the metal-line summary.

Worked examples (the converged Newton solutions of WASP-121b and HD189733b, with all four H/He spectra and the metal lines): `regression/tpm_wasp/` and `regression/tpm_hd189/`, collected in the notebook `regression/steady_solver_results.ipynb`. Figure 2 shows the four H/He spectra TPM.py saves for the WASP-121b Newton solution of `regression/tpm_wasp/` (run with `Include_He23S? True`, so the He I 10830 line is populated).

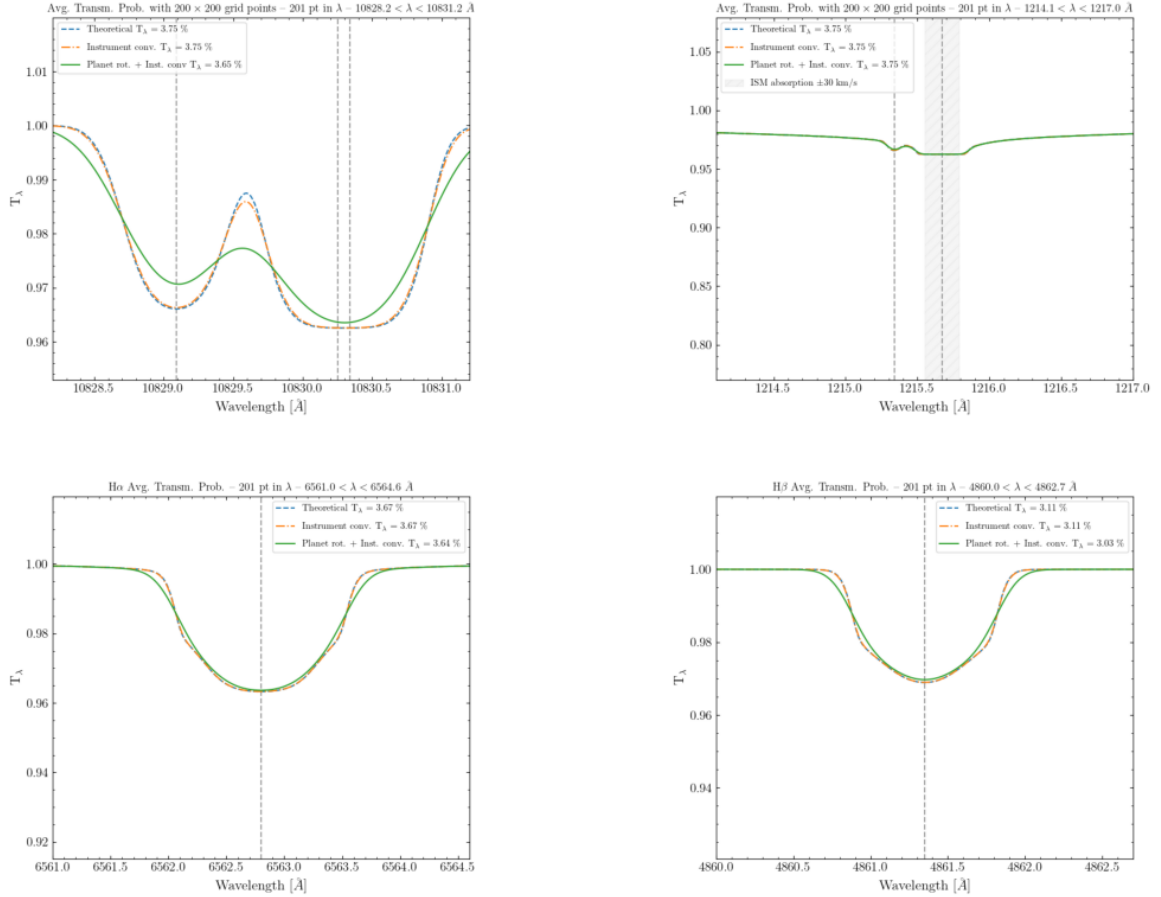


Figure 2: TPM.py output spectra for WASP-121b (Roche, Newton solution with He 2^3 S on; `regression/tpm_wasp/`): He I 10830 (top left), Ly α (top right), H α (bottom left), and H β (bottom right). Each panel shows the theoretical line, the instrument-convolved line, and the instrument+rotation-convolved line as produced by the script.

Transit-depth conventions (important). Huang et al. (2023) quote transit depths as the *effective transit radius* $R_{\text{eff}}/R_\star$, not a percent absorption; convert from the code's line absorption fraction h via $R_{\text{eff}}/R_\star = \sqrt{(R_p/R_\star)^2 + h}$. The metal NUV lines (Mg II) are quoted in a 4 Å bin, while the optical lines (Ca II, H α , Na) are line-center; line-center vs. 4 Å can differ by $\sim 3\times$ for an optically-thick line, so always compare in the same definition.

5.3 ates_io.py — named, physical-unit loaders

examples/ates_io.py loads every ATES output file into named arrays in physical units, so analysis code need not remember column orders (Table 5). The central object is the Run container: `load_run(outdir, inputfile)` fills it with the hydro profiles (r [R_p], n [cm^{-3}], v [cm s^{-1}] with a `v_kms` shortcut, p , T , `heat`, `cool`), the full ion-density dictionary `run.ion` (the 6 H/He plus 27 metal stages of §4.), and the parsed input `input.inp` in `run.inp`. By default the `_adv` (advection-corrected) files are read; pass `adv=False` for the equilibrium snapshot. Typical use:

```
import ates_io as aio
run = aio.load_run('output', 'input.inp')      # _adv profiles by default
print(aio.mdot_log10(run))                    # log10 Mdot [g/s], 2D factor incl.
print(aio.mdot_Mp_per_Gyr(run))              # same in  $M_p/\text{Gyr}$ 
xh = run.x_ion(['HI', 'HII'])                 # stage fractions of one element
plt.plot(run.r, run.T)
plt.plot(run.r, xh['HII'])
cool = aio.load_cooling('output/Cooling_breakdown.txt')
plt.semilogy(cool['r'], cool['chan']['FeII']) # one cooling channel
```

Function	Returns
<code>load_run(outdir, inputfile, adv=True)</code>	a Run: hydro profiles + <code>run.ion</code> dict + parsed input.
<code>Run.x_ion([stages])</code>	ionization fractions of one element, e.g. <code>x_ion(['HI', 'HII'])['HII'] = $n_{\text{HII}}/(n_{\text{HI}} + n_{\text{HII}})$.</code>
<code>load_hydro(path)</code>	one <code>Hydro_ioniz(_adv).txt</code> as a dict ($r, n, v, p, T, \text{heat}, \text{cool}$).
<code>load_ions(path)</code>	(r , { <code>ion</code> : density}) from <code>Ion_species(_adv).txt</code> (all 33 stages).
<code>load_cooling(path)</code>	$r, T, n_e, \text{cool_total}$ + chan dict of per-channel cooling (4 H/He channels <code>rec/coll_ion/coll_exc/brems</code> + 27 metal-line channels).
<code>load_excited_H(path)</code>	<code>Excited_H.txt</code> as a dict ($n_{2s}, n_{2p}, J_{\text{lya}}, \tau_{\text{lya}}, \text{Sproton}, \text{Hpe}, \dots$).
<code>read_input(path)</code>	<code>input.inp</code> as a dict ($R_p, RJ, M_p, MJ, a_{\text{AU}}, \text{HeH}, \text{LX}, \text{LEUV}, \dots$ + raw lines).
<code>mdot_log10(run, j_from_top=20)</code>	$\log_{10} \dot{M}$ [g/s]: $4\pi\rho v r^2$ near the outer boundary <i>including</i> the 2D approximate method factor (mirrors <code>ATES_main.f90</code> ; see §4.).
<code>mdot_Mp_per_Gyr(run)</code>	the same in units of $M_p \text{ Gyr}^{-1}$.
<code>constants</code>	$RJ, MJ, M_{\text{sun}}, \text{AU}, \text{GYR}$ (cgs).

Table 5: examples/ates_io.py public interface. Run analysis scripts either from examples/ or after adding it to the Python path (`sys.path.insert(0, '<repo>/examples')`).

The notebook examples/ATES_analysis.ipynb uses these loaders to plot the profiles and the parameter-difference comparisons of §6., and examples/make_figures.py (which generates every figure in this manual) is a further worked

example.

6. Worked examples

6.1 A first run: a generic hot Jupiter

The directory `examples/tutorial/` contains a minimal, self-contained model of a generic HD 209458b-like hot Jupiter: a spherical domain out to $10R_p$, solar metals, EUV+X-ray power-law SED. The full input `.inp` is:

```
Planet name: demoHJ
Log10 lower boundary number density [cm^-3]: 12.70
Planet radius [R_J]: 1.38
Planet mass [M_J]: 0.69
Equilibrium temperature [K]: 1320.0
Orbital distance [AU]: 0.047
Escape radius [R_p]: 2.00
He/H number ratio: 0.083
2D approximate method: Rate/2 + Mdot/2
Parent star mass [M_sun]: 1.13
Spectrum type: Power-law
Power-law index: -1.0
Use only EUV? False
[E_low,E_mid,E_high] = [ 13.60 , 123.98 , 1.24e3 ]
Log10 of X-ray luminosity [erg/s]: 27.0
Log10 of EUV luminosity [erg/s]: 29.30
Grid type: Mixed
Numerical flux: HLLC
Reconstruction scheme: PLM
Include He23S? False
Load IC? False
Do only PP: False
Force start: False
Domain mode: Spherical
Outer radius [R_p]: 10.0
du_th [PLM,WENO3]: 0.5 1.0e-3
Solver: Newton
```

The last two lines enable the automatic two-stage reconstruction and the steady-state Newton finish (§2.5), so the run converges to the true steady state without manual restarts. With `metals.inp` present (solar abundances, §3.), run `./ATES.x` in that directory. Figure 3 shows the converged profiles read back with `ates_io`: the wind heats to $\sim 10^4$ K, the velocity rises monotonically outward (transonic outflow), and hydrogen ionizes across a front near $1.2\text{--}1.4R_p$.

Tutorial run: generic hot Jupiter (spherical, H/He + metals)

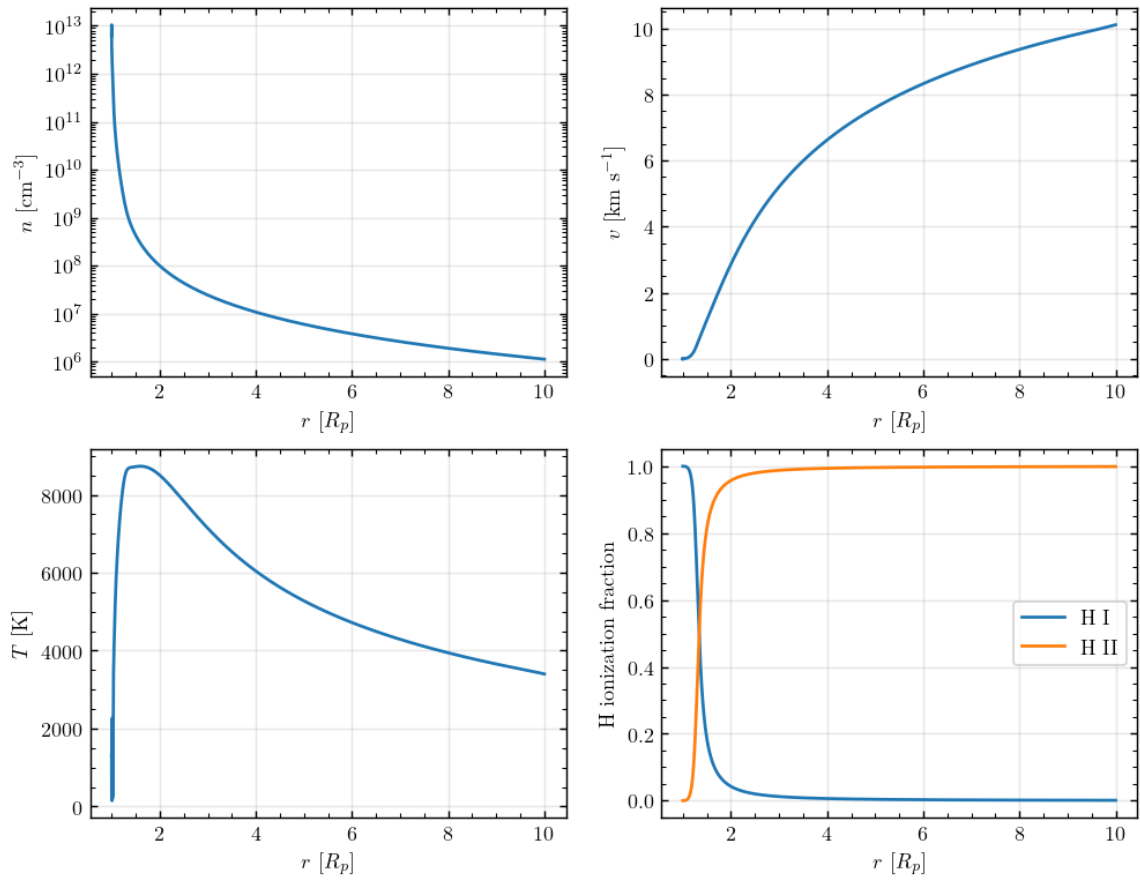


Figure 3: First-run output of the generic hot-Jupiter tutorial: density, velocity, temperature, and the H ionization fraction vs. radius.

6.2 Turning physics on and off

The same model is re-run with single-line changes to illustrate each option: *metals off* by removing `metals.inp`; *Roche/RLOF* by removing the Domain mode/Outer radius lines (the default truncates at L1); the *He 10830* line by setting `Include He23S? True`; the *H α /Balmer* physics by adding `Stellar Teff/Stellar radius`; and the *transonic IC* by adding `Transonic IC: True` for a near-unbound (deep-RLOF) base. The comparison studies below quantify the effect of the first two.

6.3 Ready-made example configurations (examples/inputs/)

For every solver/physics combination there is a ready-to-run folder under `examples/inputs/` (paths relative to the ATEs-metal root), all for the *same* planet (HD189733b) so that the effect of each option can be isolated. Each folder is self-contained: `cd` into it and run the repo-root binary (`./.././../ATES.x`); outputs land in the folder's own `output/`. Every folder differs from the baseline `01_legacy_marching/input.inp` only by the lines listed in Table 6 (so diff against the baseline shows exactly what an option adds); all nine runnable configurations have been smoke-tested.

7. Comparison studies

7.1 Effect of metal-line cooling (metals on vs. off)

Trace metals add strong line cooling (and some photoionization heating and free electrons). Figure 4 compares the tutorial planet with and without `metals.inp`: metal-line cooling lowers the peak temperature by ~ 1800 K (acting as a thermostat that prevents the H/He-only over-heating), while the H ionization structure is barely changed. The effect is far stronger for hotter, more irradiated planets (e.g. WASP-121b, where metal cooling is essential to match the observed temperatures); the detailed per-channel cooling budget is in Figure 5.

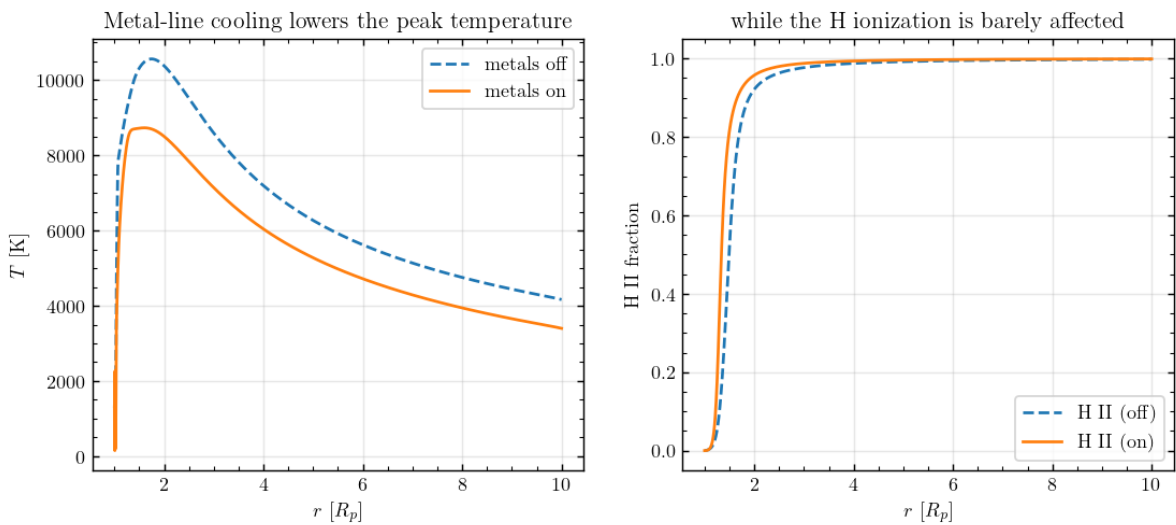


Figure 4: Metals on vs. off for the tutorial planet: metal-line cooling lowers the peak temperature (left) while the H ionization fraction is nearly unchanged (right).

examples/inputs/...	Demonstrates	Added / changed lines
01_legacy_marching	classic ATES v2 baseline: PLM marching, H/He only, Roche domain, du -based stop	(none — baseline)
02_two_stage	automatic two-stage PLM→WENO3 marching (§2.4)	du_th [PLM,WENO3]: 0.5 1.0e-3
03_newton	recommended default: two-stage warm-up + Newton finish (§2.5)	+ Solver: Newton
04_newton_from_state	Newton re-convergence of an existing state (no marching; needs output/*_IC.txt, see the folder README)	Load IC? True, Valve eps: 1.0e-4, Resid tol: 1.0e-3; run via the ATES_PTC hook
05_metals	trace-metal cooling, solar C/N/O/Mg/Ca/Na/Fe (§3.)	metals.inp present
06_he23s	He 2 ³ S metastable level (He I 10830)	Include He23S? True
07_balmer_lya	non-LTE H($n=2$) + Ly α pumping (H α /H β , §3.4)	+ Stellar Teff/radius, Deexc heat, Jlya escape-prob, Stellar Lya flux/halfwidth/boost
08_full	everything on (= the HD189733b/planet folder + Newton); feeds TPM.py	07 + metals.inp
09_spherical	spherical domain instead of the default Roche/L1 truncation (§1.3)	Domain mode: Spherical, Outer radius [R_p]: 10.0
10_warm_seed_ic	warm-seed (hot Parker overlay) initial condition	Hot Parker IC: 10000

Table 6: Ready-made example configurations under examples/inputs/ (all HD189733b). Metals on/off is a runtime switch: any folder becomes metals-on by copying a metals.inp into it.

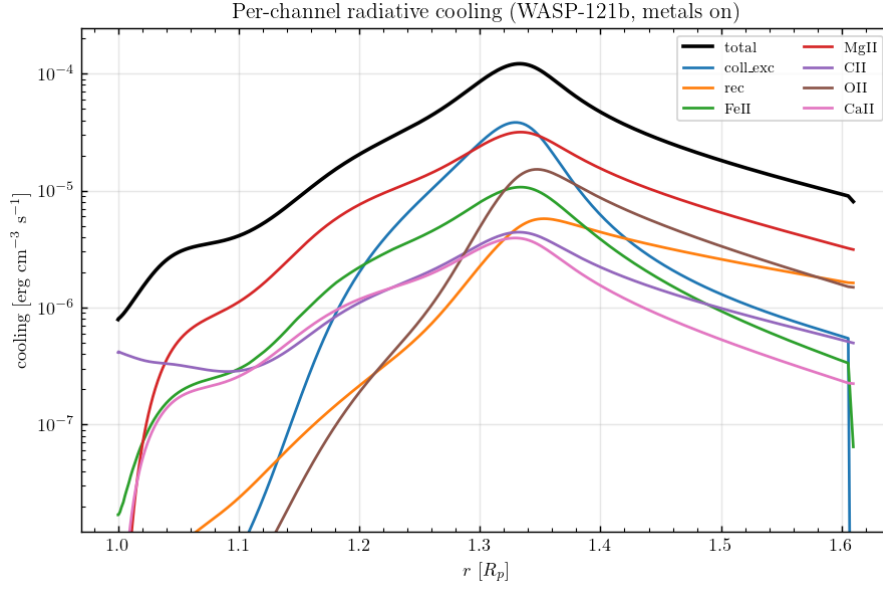


Figure 5: Per-channel radiative cooling vs. radius (WASP-121b, metals on; from `Cooling_breakdown.txt`). Below $\sim 1.4 R_p$ the metal lines (e.g. Fe II, Mg II, C/O) compete with or dominate the H/He collisional channels.

7.2 Spherical vs. Roche (RLOF) and the mass-loss rate

Removing the `Domain` mode: `Spherical` line switches on the full Roche tidal potential, truncated at the L1 radius. Figure 6 compares a spherical model (Case A) and a Roche model (Case B) of WASP-121b: the tidal potential lowers the effective gravity toward the star, so the Roche wind accelerates far faster and the mass-loss rate rises by an order of magnitude ($\dot{M}$ from ~ 0.05 to $\sim 0.34 M_p \text{ Gyr}^{-1}$ here) — the “Roche-lobe overflow” enhancement.

7.3 Transmission spectra

`TPM.py` turns a converged model into transit absorption. The metal resonance lines are the key NUV/optical diagnostics; Figure 7 compares the modelled effective transit radii R_{eff}/R_* (spherical Case A) against Huang et al. (2023). Recall the conventions of §5.2: Mg II is a 4 Å bin, Ca II/Na are line-center, and $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$.

8. Atomic and molecular data

ATES-metal is a *purely atomic/ionic* code: the modeled species are H, He (including the He 2^3S metastable triplet), and ten trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) across their first two–three ionization stages. There is **no molecular chemistry and no molecular opacity** — no H_2 , H^- , CO, or H_2O — so no molecular line lists or partition functions enter the calculation (the only “molecular” quantity in the code is the mean *molecular* weight μ , i.e. the mean mass per particle). Table 7 collects the provenance of every rate and cross section, grouped by physical process. The H/He baseline is inherited essentially unchanged from the original ATEs (Caldirola et al. 2021, 2022); the metal, excited-hydrogen, and charge-exchange data were added in this work.

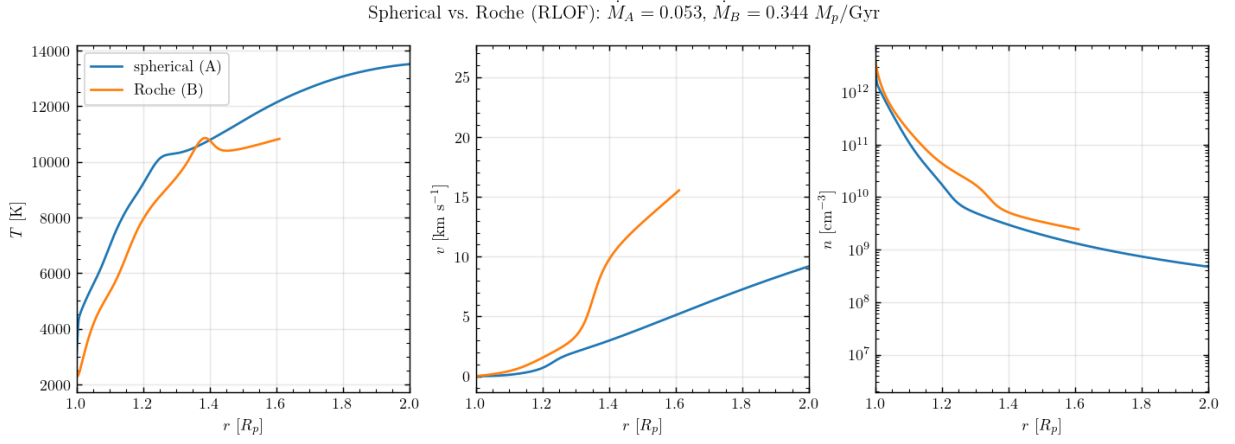


Figure 6: Spherical (Case A) vs. Roche (Case B) WASP-121b: temperature, velocity, and density. The Roche wind reaches L1 ($\approx 1.6R_p$) and accelerates much faster; $\dot{M}$ is annotated. The temperature panel is the `_adv` profile; its base is smooth because the post-processor falls back to the equilibrium solution where the flow is not a clean outflow (see the caveat in §4.). The Roche velocity (middle) dips slightly negative near the base — the genuine, physical “breathing” inflow of a near-Roche-filling planet.

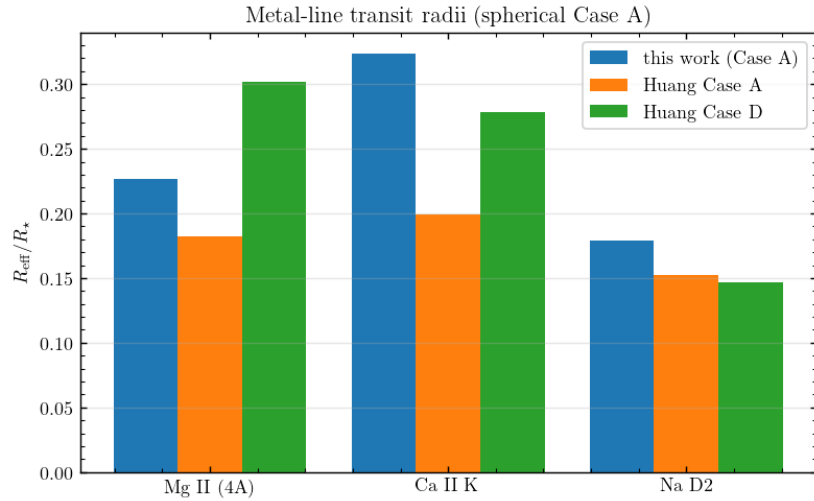


Figure 7: Metal-line transit radii R_{eff}/R_* for spherical Case A, compared with Huang et al. (2023) Case A and Case D.

Table 7: Atomic data sources, by process. Source files: `cross_sec.f90` (cross sections); `Cool_coeff.f90` (collisional rates, recombination, cooling); `charge_exchange.f90`; `excited_hydrogen.f90`; `TPM.py` (transmission line data).

Process	Species	Source
Elemental abundances		
Solar reference	10 metals (<code>metals.inp</code> template)	Asplund et al. (2009)
Helium	He/H ratio	input parameter (He/H number ratio)
Photoionization cross sections		
Bound-free $\sigma(E)$	H I	hydrogenic (Kramers) analytic form (orig. ATES)
	He I	analytic fit (orig. ATES)
	He I 2^3S (metastable)	Norcross (1971) fit
	C, N, O, Mg, Si, Na, Fe (+ ions)	Verner, Ferland, Korista & Yakovlev (1996), single-shell fits
	K I	Verner & Yakovlev (1995)
Opacity model (runtime)	all (via <code>opacity.inp</code>)	A=analytic above; P=pressure-broadened (Robinson & Catling); T=tabulated files
Collisional (electron-impact) ionization		
Rate $C(T)$	H I, He I, He II	analytic fits (orig. ATES)
	C–Fe, all stages	Voronov (1997, ADNDT 65, 1)
Recombination (radiative RR + dielectronic DR)		
Case B (RR)	H II, He II, He III	Hui & Gnedin (1997) fits (DR negligible)
Metastable	He II $\rightarrow 2^3S, 1^1S$	power-law fits (orig. ATES)
RR + DR	C, N, O, Mg, Si, Na, K, S, Ca II (+ stages)	Badnell (2006, ApJS 167, 334) RR + Badnell adf48 DR (TAMOC clist_K)
RR only	Ca I (Ca II+e)	Shull & Van Steenberg (1982); no DR
RR + DR	Fe I, Fe II	Huang et al. (2023) analytic fit (validation; Badnell switch deferred)
Charge exchange (charge transfer)		
metal + H/H ⁺ (default)	23 reactions (Group A)	Huang et al. (2023, ApJ 951, 123) Table 4
full network (<code>cx_full</code>)	+ He+H, metal+He, metal+metal	Huang et al. (2023) Table 4 (Groups B/C/D)
Radiative cooling		
recomb., coll. ion./exc., free–free	H, He	orig. ATES (Hui & Gnedin 1997; Black 1981; Gaunt-factor bremsstrahlung)

Table 7, continued

Process	Species	Source
metal line cooling (analytic)	C, O (+ ions), etc.	AIOLOS fits (ported; Schulik & Booth 2023, Cloudy-calibrated)
metal line cooling (tabulated)	Mg I/II, Ca II H&K, Na I D, Fe II	CHIANTI v11.0.2 (Burgess–Tully descaled); Na I D via effective $\bar{g} \approx 0.2$
fine-structure (opt., use_2lev_cool)	[O I] etc.	Lique et al. (2017) / Draine (2011)
Excited hydrogen $H(n=2)$ and $Ly\alpha$		
$2s/2p$ rate equilibrium	$H(n=2)$	Christie, Arras & Li (2013) Table 2
recombination cascade to $n=2$	H	Draine (2011)
$Ly\alpha$ pumping field (mode 0)	H	Huang et al. (2017) Eq. 6
Transmission line data (TPM.py)		
λ , f -value, A_{ki}	He 10830, $Ly\alpha$, $H\alpha$, $H\beta$, Mg II, Ca II H&K, Na I D	NIST ASD
line profile	all	Voigt (Faddeeva, <code>scipy.special.wofz</code>)

Note on Fe. Iron recombination currently uses the Huang et al. (2023) analytic RR+DR fit (chosen to validate against Huang’s Fig. 12) while every other metal uses the Badnell tabulation; unifying Fe onto Badnell is a documented deferred item. **Note on completeness.** The metal grid carries the stages needed for the XUV-driven ionization balance and the observed NUV/optical lines; higher stages and additional elements (e.g. the full Fe forest, K I, Mg I) beyond those tabulated above are not modeled.